

Evaluating Foundation Model Representations for Cell Segmentation and Classification in Microscopy Images

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Introduction

Foundation models and large-scale pretraining have become central to visual representation learning, including in computational pathology. Models trained on natural images or large microscopy and histopathology datasets are increasingly reused as fixed feature extractors for downstream tasks such as diagnosis, disease sub-typing, biomarker quantification, treatment response estimation, and survival prediction [1, 2, 3]. While this paradigm offers strong performance and reduced training costs, recent work suggests that substantial performance gaps remain between general-purpose representations and domain-specific embeddings when applied to fine-grained cellular tasks [4]. In particular, cell segmentation and cell classification place strong demands on spatial precision and domain-specific visual cues [5]. This project is motivated by the need to better understand these gaps and to systematically analyse how patch-level embeddings transfer across datasets and tissue types for cell segmentation and classification in microscopy images, particularly in histopathology.

Objectives and Methodology

Students will first conduct a literature review on foundation models for microscopy images, with particular emphasis on histopathology. The review should cover pretraining strategies (supervised vs. self-supervised), dataset scale and domain specificity, and architectural design choices. Building on these insights, students will implement an experimental study in Python to systematically evaluate different foundation models as encoders within a UNETR architecture [6] for cell segmentation and cell classification. Models will be assessed both as frozen feature extractors and, where appropriate, under end-to-end fine-tuning, enabling direct comparison of these regimes. Experiments will analyse how factors such as pretraining domain, pretraining strategy, and representation design choices influence performance, with the goal of identifying limitations of current models and proposing directions for improving cellular-level modelling.

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References

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